

Database Management Systems and Design

SEIS 630 -Spring 2026

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Class Meeting*

SEIS 630-1 Monday
5:30pm-8:30pm
OSS 329/Zoom
SEIS 630-02 Wednesday
5:30pm-8:30pm
OSS 329/Zoom

Office Hours*

Monday/Wednesday: 4:00-5:00pm
*Heads-up are good
*By appointment also welcome
*Zoom is default (join to start and
be patient if you don't have an
appointment)

1 Course Description

This course provides a broad introduction to the subject of database management systems (DBMS) and practical experience using database systems. The primary focus is on relational database systems (RDBMS) though the course will examine other non-relational systems.

2 Objectives

1. Gaining a basic understanding of the subject (e.g., factual knowledge, methods, principles, generalizations, theories) [IDEA Survey Question 1]
 - Historical context; modeling data attributes and relationships conceptually; understanding the syntax, semantics, and variations in Structured Query Language (SQL); algorithms and data structures involved in databases;
2. Learning to apply course material (to improve thinking, problem solving, and decisions) [IDEA Question 3]
 - Solving data organization issues; resolving potential data anomalies by normalization; performance reasons for denormalization; implementing and querying data models in databases;
3. Developing specific skills, competencies, and points of view needed by professionals in the field most closely related to this course [Idea Question 4]
 - Installing software; becoming familiar with SQL; communicating data models through diagrams; learning about different DBMS products and evaluation criteria; database-backed application designs; transaction control and security; scaling and availability;

3 Prerequisites

There are no prerequisites for this class. However, basic computer skills like typing, installation of programs, troubleshooting errors, and googling are assumed.

4 Course Materials

4.1 Textbook

The textbook of the course is:

- Database Design, Query, Formulation, and Administration: Using Oracle and PostgreSQL, Eighth Edition, by Michael Mannino. (ISBN: 978-1948426954) (UST bookstore, Amazon.com, publisher)

We may also have other readings if time permits.

Note: for readings, it is usually not necessary to read the chapters completely. Prioritization is key. The intro and summary sections are what you should read first. Be critical and choose to focus or skim depending on your situation. For example, details about databases that we aren't using in class may be skipped on first reading if this is your first experience with databases. However, if you have used a given database, you may want to read about it to help connect the topics to your past experiences.

4.2 Software

We will primarily use Oracle tools to learn SQL and to make Entity Relationship diagrams (ER diagrams). We will start using Oracle SQL Developer, which is a client that we will use to connect to the class Oracle Cloud database. We will use Oracle Data Modeler for ER diagrams and we will use an Oracle Cloud account to create databases and access them Oracle SQL Developer client. Other software may be demonstrated in for wider context or as a backup in the case of technical issues/breaking changes. For the final project, you will have the option to explore other database software. . Oracle Live SQL offers an online interface to an Oracle Cloud database that is another option that runs in a browser and provides some tutorials, but data and schemas are not saved between sessions.

Later in the semester we may use additional software, such as alternative DBMS's and using Python to access data programmatically.

4.3 Canvas

Lecture slides, homework assignments, and project information will be posted on `canvas.stthomas.edu`.

5 Grading

Your course grade will be based on your performance in the following components:

Requirement	% of Grade
1. Assignments	30%
2. Quizzes	30%
3. Project	30%
5. Participation	10%

Assignments are low stakes in terms of grading, which means that you will you may have opportunities to revise submissions based on in-class discussions in order to learn from any mistakes or uncertainties. Some assignments are also partly open-ended and may involve submitting a write-up. The goal for this is to prepare and ramp up for the project.

Quizzes are higher stakes assessments of knowledge that are intended to test your understanding of concepts, terminology, and SQL commands. There's no mid-term or final exam, but the quizzes aim to have a similar goal of assessment.

The project will be on a topic of your choosing and will involve either designing a database around scenario you choose or researching a database-related topic. There will be several deliverables, feedback activities, a writeup, and a brief video recording. The goal of the final project is to allow students to customize their learning goals and accommodate different levels of prior knowledge about databases.

5.1 Grade Scale

The percentages apply to your overall score in the class. In accordance with departmental grading policies, there is no "D" grade. To compensate for variability in the homework, exams, and projects, the scores may be curved, reweighted, or extra credit may be offered.

Grade	Point Range
A	$\leq 93\%$
A-	[90%, 93%)
B+	[87%, 90%)
B	[83%, 87%)
B-	[80%, 83%)
C+	[77%, 80%)
C	[73%, 77%)
C-	[65%, 73%)
F	$< 65\%$

5.2 Class Participation

Class participation is encouraged. Participation includes listening as well as talking. Quantity is not a substitute for quality. Participation also includes online discussions, some of which will be graded explicitly as a component of homeworks/project.

On Zoom, participation includes having working audio and video. Video should be enabled when there are breakout rooms and the professor prefers having the video enabled during to be able to see student engagement.

Baseline participation includes attendance and coming on time. Sign into Zoom 5-10 minutes early to take attendance and work out connection issues. Acceptable reasons to turn off video include using the bathroom, picking one's nose, stretching, etc., and network connection issues. Let the professor know if there is network connection issues that prevent use of video in Zoom. It is preferable to use voice to communicate in Zoom, so please unmute yourself and barge in unless instructed otherwise. Turning the video and mic on is also a good signal that you want to speak. The Zoom chat feature is preferred as a back-channel. For example, in a normal classroom, you might whisper to your classmates. The chat in Zoom is like that, except that chat is not disruptive and in fact encouraged. The professor will monitor Zoom chat but may not see messages immediately. At the end of class in Zoom, we will unmute and make a toast/cheer to say goodbye. This was intended to build camaraderie in these times of social distancing and we will continue it even though sentiment has improved post-covid. Feel free to show the class a beverage/snack, pets, kids, or Zoom background.

6 Class Schedule

For more info, see the GPS semester dates and deadlines page. **Note:** currently only the Monday class has a break for Easter (Monday). To keep the two sections aligned will discuss adding a break for Wednesday or removing one for Monday.

Mon.	Wed.	Topics
2/2	2/4	Lecture 1: (prep: read ch. 1, 2) introductions and intro survey, class organization, class overview, SQL and ER diagrams overview, database history, database roles/jobs; hands on: finding textbook data, registering for Oracle Live SQL, intro to data import, homework: get textbook, sign up for Live SQL, download textbook data, intro survey.
2/9	42/11	Lecture 2: (prep: ch. 4,1-2 if necessary) intro to SQL, SELECT and DML/DQL; hands on: ch. 4 queries using SQL Developer/Oracle Cloud
2/16	2/18	Lecture 3: (prep: ch. 3) normalization and the relational model, relational algebra, relations vs. relationship in ER diagrams, shands on: loading survey data to Live SQL; quiz 1: SELECT, chapters 1 and 2
2/23	2/25	Lecture 4: (prep: ch. 7, optional: Tidy Data paper) data modeling/conceptual schema and ER diagrams, maximum and minimum cardinality, strong/weak entities; assignment 2: practical normalization of survey data using INSERT and SELECT statements
3/2	3/4	Lecture 5: (prep: ch. 5) database design/logical schema, keys, common design patterns, default values, object relational mappings, final project ideas; hands on: installing Oracle Data Modeler and loading DDL (forward engineering) and exporting DDL (reverse engineering); Note: prof. traveling Mar. 5-8
3/9	3/11	Lecture 6: (prep: ch. 6) systems development life cycle (SDLC), Database construction/DDDL, CREATE TABLE, constraints, views, indexes, INSERT/UPDATE/MERGE, user defined functions, triggers, and persistent stored modules, hands on: continuing Data Modeler;
3/16	3/18	Lecture 7: (prep: ch. 8) Storage, page files, log files, B-Trees, bitmap indexes, inverted indexes/full text search, clustered indexes
3/23	3/25	Lecture 8: (prep: ch. 16 and 17) Multiuser databases, three layer application design, concurrency, security, backup, recovery, administration, hands on: Migration example in Django.
3/30	4/1	Lecture 9: (prep: ch. 9, 10, and 11) Advanced queries, inner and outer joins, nested subqueries, correlated subqueries, views, null values, hierarchical and recursive queries, user-defined functions, triggers, stored procedures. Note: Easter Apr. 5, Easter Monday Apr.6
4/13	4/8	Lecture 10: (prep: ch. 18) Distribution of DBs, desktop vs. grid vs. cloud, DB-backed web applications, OSI model, DB drivers, thin/rich clients, model-view-controler framework, HTTP/REST/CRUD, web stacks, hands on: creating a database design and populating it with example data
4/20	4/15	Lecture 11: (prep: ch 12, 13, 14, and 15) Big data, DBAs and ethics, data warehouses, star and snowflake schemas, extract-transform-load (ETL), distributed databases, cloud DBs, serialization and storage issues, map-reduce and search, hands on: continued from previous week; homework: submit draft
4/27	4/22	Lecture 12: NoSQL/Non-relational databases; follow ups to spend more time on topics if necessary;
5/4	4/29	Lecture 13: Final project draft due, group feedback activity; homework: group feedback, final writeup presentation/video Note: Thanksgiving?
5/11	5/6	Lecture 14: Final Project Presentations

7 Policies

7.1 Email

I will reply within 24 hours during weekdays and may take longer on weekends. If I don't reply in that time period, please don't be afraid to follow up.

7.2 Attendance

Whether in-person or on Zoom, please come 5-10 minutes early to have time to get a seat or deal with any connection issues. More than two absences may result in deduction to grade. Please email the professor as soon as possible about any lateness or absences.

7.3 Late Policy

The professor reserves the discretion whether to accept late assignments. Late submissions will be subject to deductions and/or zero credit, but it is still beneficial for learning to complete and hand in the late work, so please hand in late work.

Due dates on Canvas specify times as well as dates. The professor may offer a grace period if the submission is overdue by only a few minutes/hours. For example, if the assigned due date is midnight but the submission time is 1:30am, this would most likely be fine. It is a good idea to communicate planned lateness if possible.

7.4 Time/Effort Budget

There will be approximately 3 hours of classroom time per week. Students should budget approximately 2-3 times the classroom time for outside-the-classroom study time, such as homework, reading, online activities, test preparation, email, etc. This will vary due to varied student ability and background knowledge as well as due to course schedule.

7.5 Academic Integrity

Academic integrity is defined as not cheating and not plagiarizing, and more positively as honesty and trust. Honesty and trust among students and between students and faculty are essential for a strong, functioning academic community. Consequently, students are expected to do their own work on all academic assignments, tests, projects and research/term papers unless instructed otherwise. Academic dishonesty, whether cheating, plagiarism or some other form of dishonest conduct related to academic coursework and listed in the Student Policy Book under "Discipline: Rules of Conduct" will automatically result in failure for the work involved. But academic dishonesty could also result in failure for the course and, in the event of a second incident of academic dishonesty, suspension from the University. For more info see the Appendix in Section 8 below.

7.6 Statement Related to Influenza and Covid-19

The University of St. Thomas is committed to a healthy campus community. To help limit the spread of influenza, Covid-19, and other illnesses, the Center for Disease Control has provided college campuses the following recommendation: students, faculty, or staff with influenza-like illnesses (temperature of 100.0 or greater, plus a cough or sore throat) are directed to self-isolate (or stay home) for at least 24 hours after their fever is gone without the use of fever-reducing medicine (in the case of general illnesses) or 10 days in the case of Covid-19. In the event that students are unable to attend classes due to this self-isolation recommendation, they should inform the professor of their absence.

7.7 Students with Disabilities

Academic accommodations will be provided for qualified students with documented disabilities including but not limited to mental health diagnoses, learning disabilities, Autism, Attention Deficit Disorder, chronic medical conditions, visual, mobility, and hearing disabilities. Students are invited to contact the Disability Resources office about accommodations early in the semester. Appointments can be made by calling 651-962-6315 or in person in Murray Herrick, room 110. For further information, you can locate the Disability Resources office on the web at <http://www.stthomas.edu/enhancementprog/>.

7.8 Diversity

St. Thomas is committed to diversity in many forms. Besides the typical diversity of outer appearance, students, faculty, and staff may have different cultural, intellectual, and economic backgrounds. Aiming to support diversity in the classroom can be seen as a positive benefit, like creating a diversified portfolio in a financial investment. It is preferable to strive for this positive, beneficial view of diversity where everyone respects and learns from each other and the organization grows sustainably and earns dividends from the investment in diversity. However, the university also provides services for addressing times when this is not the case. The Dean of Students Office provides services for responding to instances of bias or hate, financial hardship, and sexual harassment. If you have experienced or witnessed bias, hate, or sexual misconduct or if you don't have access to sufficient food or a safe, stable place to live, please contact the Dean of Students Office by phone at 1(651) 962-6050 or in person at room 241 in the Anderson Student Center.

7.9 Recording of Classroom Activities

Recordings of class sessions using any device must be authorized with permission from the professor. Recordings will be provided under "Zoom Pro->Cloud Recordings" in Canvas.

8 Appendix: Academic Integrity

8.1 AI Policy

The professor will explain for which coursework it is appropriate to use AI.

In general, think of learning as an exercise and try to work through obstacles and resistance on your own. If you use AI without thought, resistance, and struggle, you will not exercise your brain as a muscle, you won't learn, and you will become mentally weak. Try to use AI as a last resort rather than a crutch.

That being said, AI is another technical tool that promises/threatens to change many things in professional technical work. So you should aim to be familiar with AI's strengths and weaknesses, and prepare yourself for AI being a part of future professional work.

To balance these competing concerns and to fit AI into academic practices, especially academic integrity, please cite the use of AI when you use it in class. An appropriate citation is a link to the conversation you used as a part of an assignment, quiz, or project.

8.2 Exam Proctoring Policy

The University of St. Thomas continues to strive for academic excellence and integrity in times of continuous technological advancement. The University has selected third-party online exam monitoring tools ("tools") to assist in combating academic dishonesty in the online testing environment. This is part of a three-step approach of prevention, detection, and communication. By registering for a course within the School of Engineering's Graduate Programs in Engineering and Software, students are agreeing to install the tools on exam-taking devices for the duration of any course exam, if required by the faculty instructor.

8.3 Academic Integrity Policy

Honesty and trust among students and between students and faculty are essential for a strong, functioning academic community. Students are expected to complete all academic work on their own. Academic dishonesty is not tolerated in the School of Engineering at the University of St. Thomas.

Students are required to complete/submit all academic work as one's own. If a student fails to do so, a grade of zero is issued. If the student commits a second academic dishonesty violation in the same or future course, a course grade of F is issued. If a third violation occurs, the student is dismissed from the program.

*Note on ChatGPT and other large language models (LLMs): if you use ChatGPT or other LLM or generative AI technology, you are expected to cite it like any other external sources you use.

Here are the common ways to violate academic integrity:

Cheating Using, accessing, or trying to use unauthorized materials, information, or study aids in any academic work submitted for credit; giving or receiving unauthorized materials or aid in an academic exercise.

Fabricating/misrepresenting Falsifying or inventing any information or citation in an academic exercise; fabricating or misrepresenting a reason for an absence or missed deadline; submitting academic work completed largely in one class to satisfy an assignment in a second class without instructor permission; fabricating or tampering with academic records.

Facilitating Academic Dishonesty Knowingly helping or attempting to help another student commit academic dishonesty.

Plagiarism Claiming or submitting the ideas, words, or statements of another person as one's own without proper citation regardless of intent to deceive. Even inadvertent plagiarism is a serious matter; students must familiarize themselves with citation standards.

Cheating

In cases of cheating, the instructor will impose a minimum sanction of failure of work involved. The instructor will inform the student and the program director in writing of: 1. the nature of the offense, 2. the penalty imposed within the course, 3. the recommendation of the instructor as to whether further disciplinary action by the director is warranted.

If the instructor or the director of the program determines that further disciplinary action is warranted, a disciplinary hearing shall be commenced at the request of either the instructor or the director. (If there is a previous offense of this nature on the student's record, a hearing is mandatory.)

Plagiarism

Here are examples of various kinds of plagiarism.

In each instance, the source is a passage from p.102 of E.R. Dodd's *The Greek and the Irrational* (Berkeley, 1971; reprinted: Boston: Beacon, 1957). First here is the original note, copied accurately from the book *Functions*, Dodds 12, p. 102:

If the waking world has certain advantages of solidity and continuity its social opportunities are terribly restricted. In it we need as a rule, only the neighbors, whereas the dream world offers the chance of intercourse, however fugitive, with our distant friends, our dead, and gods. For normal men it is the sole experience in which they escape the offensive and incomprehensible bondage of time and space.

Here are five ways of plagiarizing this source: (If you have any questions about plagiarism ask the instructor)

Word-for-word continuous copying without quotation marks or mention of the author's name (copying images as well).

Dreams help us satisfy another important psychic need—our need to vary our social life. This need is regularly thwarted in our waking moments. If the waking world has certain advantages of solidity and continuity, its social opportunities are terribly restricted. In it we need, as a rule, only the neighbors, whereas the dream world offers the change of intercourse, however fugitive, with our distant friends, our dead, and our gods. We awaken from such encounters feeling refreshed, the dream having liberated us from the here and now...

Copying many words and phrases without quotation marks or mention of the author's name.

Dreams help us satisfy another important psychic need—our need to vary our social life. In the waking world our social opportunities, for example, are terribly restricted. As a rule, we usually encounter only the neighbors. In the dream world, on the other hand, we have the chance of meeting our distant friends. For most of us it is the sole experience in which we escape the bondage of time and space....

Copying an occasional key word or phrase without quotation marks or mention of the author's name.

Dreams help us satisfy another important psychic need--our need to vary our social life. During our waking hours our social opportunities are terribly restricted. We see only the people next door and our business associates. In contrast, whenever we dream, we can see our distant friends. Even though the encounter is brief, we awaken refreshed, having freed ourselves from the bondage of the here and now...

Paraphrasing without mention of the author's name.

Dreams help us satisfy another important psychic need--our need to vary our social life. When awake, we are creatures of this time and this place. Those we meet are usually those we live near and work with. When dreaming, on the other hand, we can meet far-off friends. We awaken refreshed by our flight from the here and now.

Taking the author's idea without acknowledging the source.

Dreams help us to satisfy another important psychic need—the need for a change. They liberate us from the here and now, taking us out of the world we normally live in....

If you quote anything at all, even a phrase, you must put quotation marks around it, or set it off from your text; if you summarize or paraphrase an author's words, you must clearly indicate where the summary or paraphrase begins and ends; if you use an author's idea, you must say that you are doing so. In every instance, you also must formally acknowledge the written source from which you took the material. (This includes material taken from the World Wide Web and other Internet sources.) Reprinted from "Writing: A College Handbook" by James A.W. Herrerman and John E. Lincoln. By Permission W.W. Norton & Co. Inc., Copyright 1982 by W.W. Norton & Co. Inc. Students are encouraged to report incidents of academic dishonesty to course instructors.

When academic dishonesty occurs, the following procedures will be followed.

A. The instructor will impose a minimum sanction of failure for the work involved. The instructor will notify the student and the appropriate academic dean/director in writing of the nature of the offense and that the minimum sanction has been imposed. The instructor may recommend to the dean that further penalties should be imposed. If further penalties are imposed, the dean/director will notify the student immediately and the student will have five working days to respond to the intention to impose additional penalties. The student has the right to respond to the charge of academic dishonesty and may request in writing that the dean review the charge of academic dishonesty as fully as possible. If the dean/director determines that no further sanctions will be applied, the instructor's sanction will stand and the instructor's letter to the dean/director and student will be placed in the student's file. If no further charges of academic dishonesty involving the student occur during the student's tenure at St. Thomas, the materials will be removed from the file upon graduation.

B. If the student has been involved in a previous incident of academic dishonesty, the dean will convene a hearing, following guidelines listed under "Hearings and Procedures" in the Student Policy Book. During the hearing, all violations of academic integrity will be reviewed. The student and the faculty member charging the most recent incident will be present at the hearing.

C. In either situation, A or B, if the dean/director determines that further sanctions are warranted, the student will be informed in writing. Among the sanctions considered by the dean/director will be the following: failure for the course in which the incident occurred; suspension from the university for the following semester; expulsion from the university; community service; a written assignment in which the student explores the principles of honesty and trust; other appropriate action or sanctions listed under "Sanctions" in the Student Policy Book. The materials relating to the incident including the instructor's original letter to the student and dean and the dean's decision following the hearing, will become part of the student's file.